



Faculty of Computer Applications

**BCA Sem-1
(AY-2026-27)**

Lab Book

**Sub Code : 05BC3103
(Programming Practices : 1)**

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GitHub Link | <https://github.com/reaandalpautra-pixel/BCA--PP-1---92600527028-Mo.Reaan-Dalpautra>

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Index

Unit : 2	
Program No	Definition
1.	Write a Program to Print "Hello World".
2.	Write a Program that takes 2 Values from user and Perform All Arithmetic Operation.
3.	Write a program that takes 3 values for Principle Amount, Rate of Interest and No. of Years and find out Simple Interest.
4.	Write a program that input value from user and find out Area of Circle.
5.	Write a program that input two values from user and find out Area of Rectangle.
6.	Write a program that input 3 values from user and find out its Average.
7.	Write a program to Calculate the Square and Cube of a number.
8.	Write a program Interchange the value of two numbers using third variable.
9.	Write a program Interchange the value of two numbers without using third variable.
10.	Write a program that input quantity, price, discount and display final amount.

(1) Write a Program to Print "Hello World".

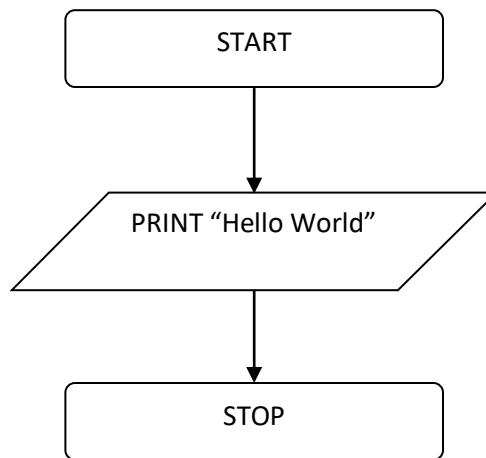
Algorithm :

Step 1 : START

Step 2 : PRINT "Hello World"

Step 3 : STOP

Flowchart :

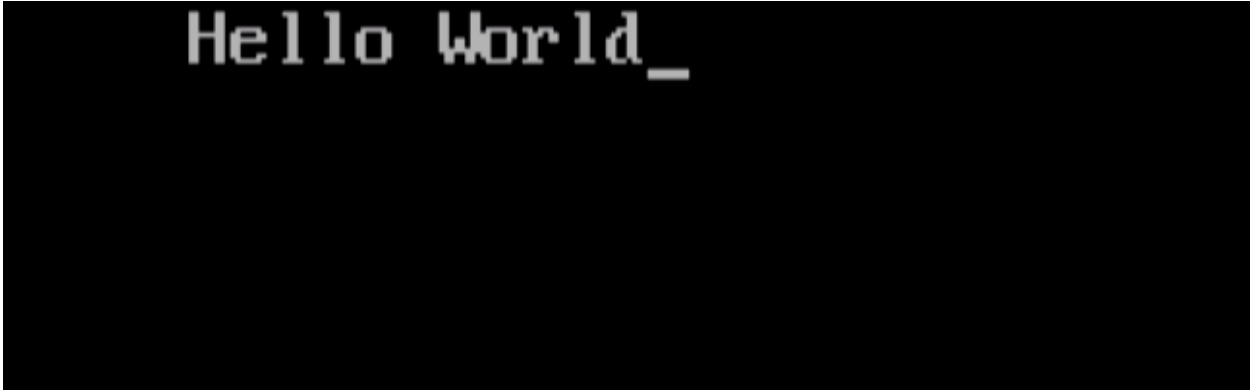


Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
    clrscr();
    printf("Hello World");
    getch();
}
```

Output :

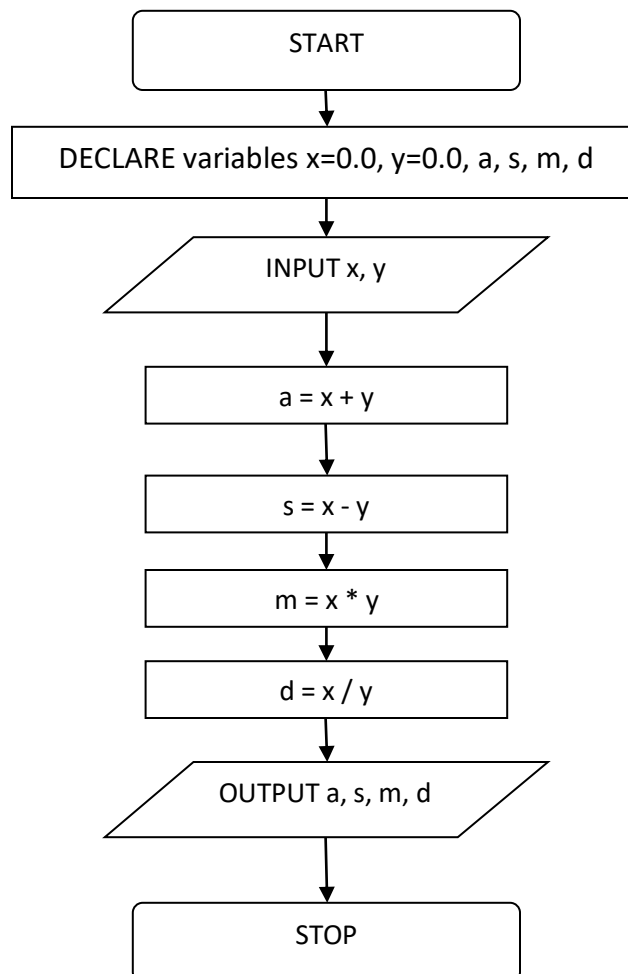
A screenshot of a terminal window with a black background. The text "Hello World_" is displayed in a light gray, monospaced font. The text is positioned in the upper left area of the terminal, followed by a cursor underscore.

(2) Write a Program that takes 2 Values from user and Perform All Arithmetic Operation.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables $x=0.0$, $y=0.0$, a , s , m , d
Step 3 : INPUT x, y
Step 4 : $a = x + y$
Step 5 : $s = x - y$
Step 6 : $m = x * y$
Step 7 : $d = x / y$
Step 8 : OUTPUT a, s, m, d
Step 9 : STOP

Flowchart :



Code :

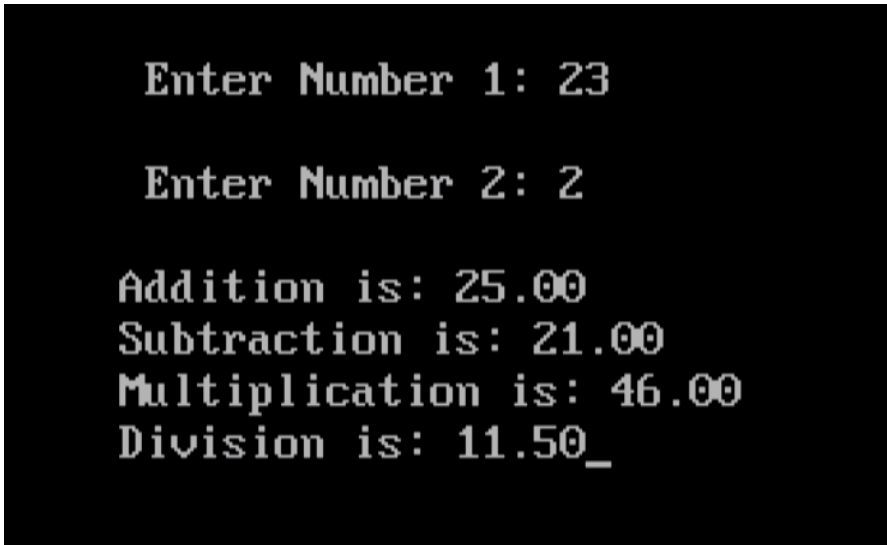
```
#include<stdio.h>
#include<conio.h>

void main()
{
    float x,y;
    clrscr();
    printf("\n Enter Number 1: ");
    scanf("%f",&x);
    printf("\n Enter Number 2: ");
    scanf("%f",&y);

    printf("Addition Is: %.2f \n",x+y);
    printf("Subtraction Is: %.2f \n",x-y);
    printf("Multiplication Is: %.2f \n",x*y);
    printf("Division Is: %.2f \n",x/y);

    getch();
}
```

Output :

A screenshot of a terminal window with a black background and white text. The text shows the output of a C program. It starts with 'Enter Number 1: 23' and 'Enter Number 2: 2'. Then it shows the results of four operations: 'Addition is: 25.00', 'Subtraction is: 21.00', 'Multiplication is: 46.00', and 'Division is: 11.50_'.

```
Enter Number 1: 23

Enter Number 2: 2

Addition is: 25.00
Subtraction is: 21.00
Multiplication is: 46.00
Division is: 11.50_
```

(3) Write a program that takes 3 values for Principle Amount, Rate of Interest and No. of Years and find out Simple Interest.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables SI, P, R, T

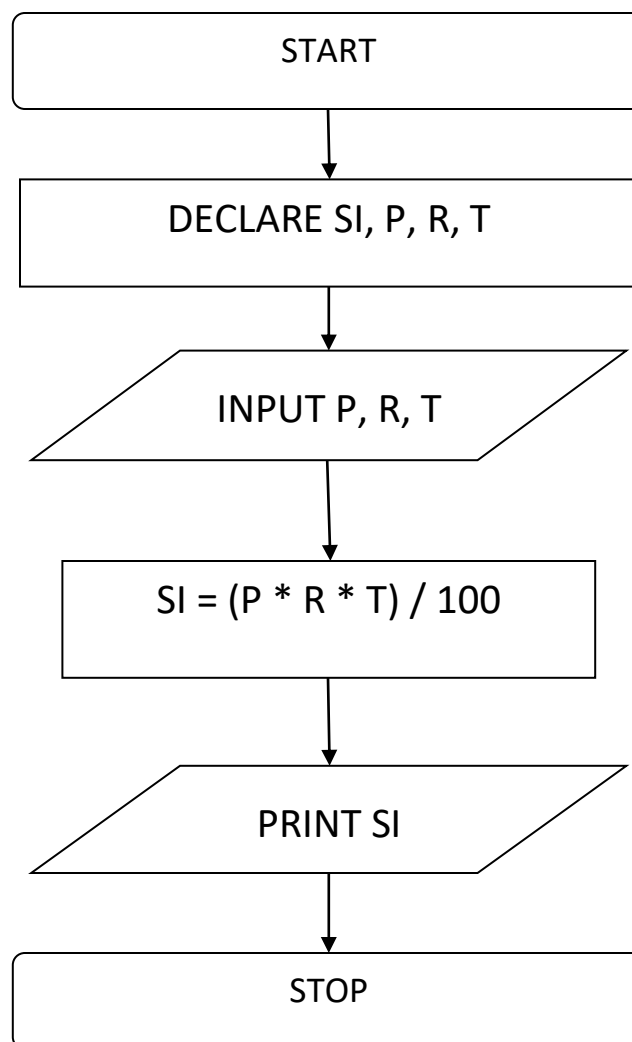
Step 3 : INPUT P, R, T

Step 4 : $SI = (P * R * T) / 100$

Step 5 : PRINT SI

Step 6 : STOP

Flowchart :



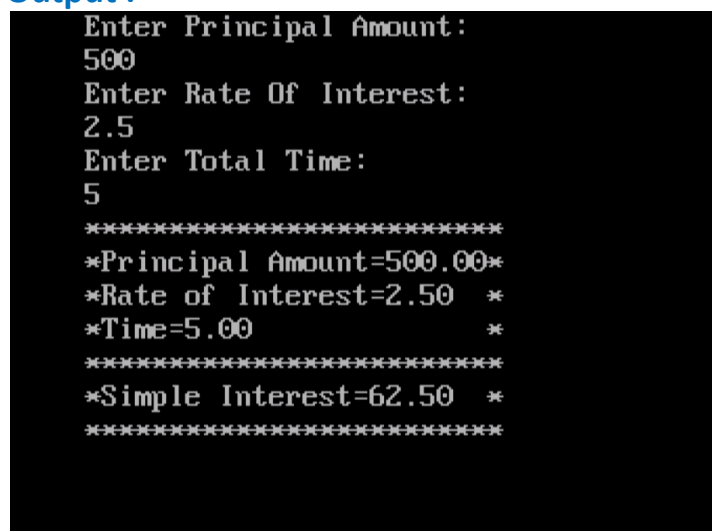
Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
    float SI,P,R,T;
    clrscr();
    printf("Enter Principal Amount: \n");
    scanf("%f",&P);
    printf("Enter Rate Of Interest: \n");
    scanf("%f",&R);
    printf("Enter Total Time: \n");
    scanf("%f",&T);
    SI=(P*R*T)/100;
    printf("*****\n");
    printf("*Principal Amount=%.2f*\n",P);
    printf("*Rate of Interest=%.2f *\n",R);
    printf("*Time=%.2f *\n",T);
    printf("*****\n");
    printf("*Simple Interest=%.2f *\n",SI);
    printf("*****");

    getch();
}
```

Output :



```
Enter Principal Amount:
500
Enter Rate Of Interest:
2.5
Enter Total Time:
5
*****
*Principal Amount=500.00*
*Rate of Interest=2.50 *
*Time=5.00 *
*****
*Simple Interest=62.50 *
*****
```

(4) Write a program that input value from user and find out Area of Circle.

Algorithm :

Step 1 : START

Step 2 : DECLARE variable r, area

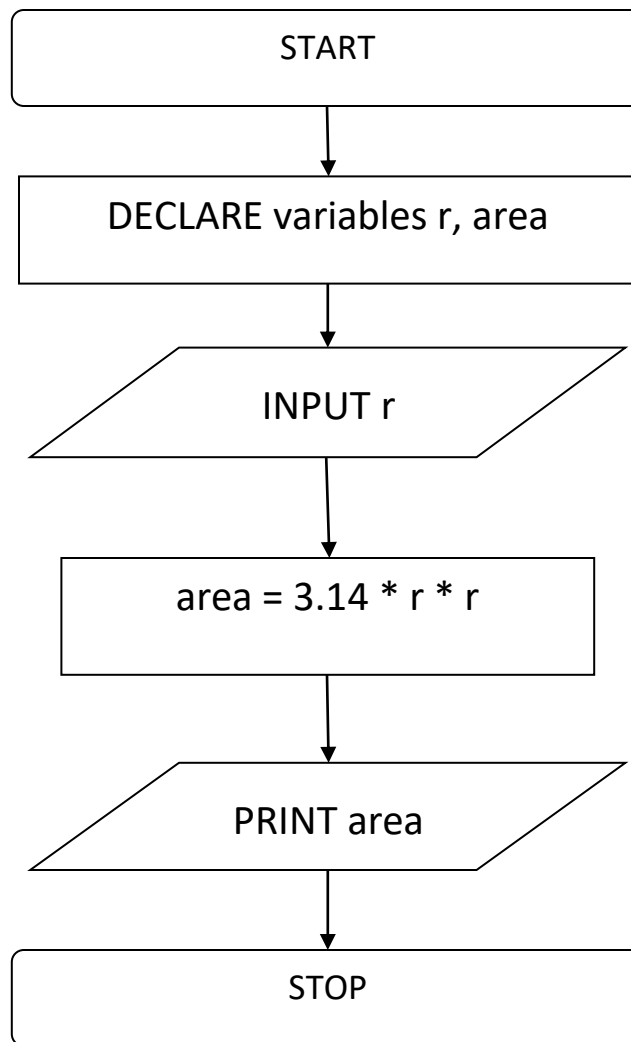
Step 3 : INPUT r

Step 4 : $\text{area} = 3.14 * r * r$

Step 5 : PRINT area

Step 6 : STOP

Flowchart :

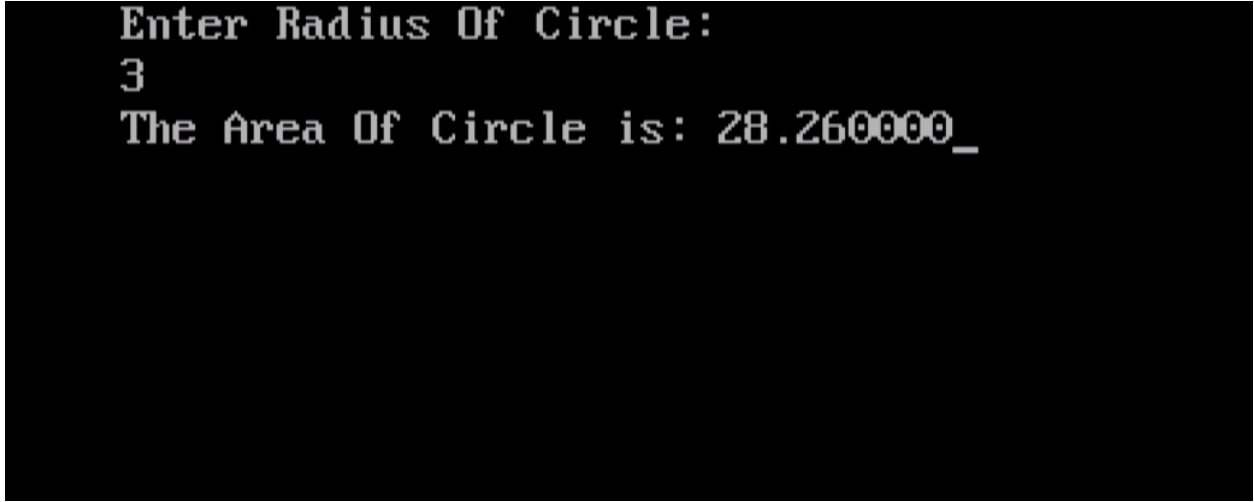


Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    float Radius;
    clrscr();
    printf("Enter Radius Of Circle: \n");
    scanf("%f",&Radius);
    printf("The Area Of Circle is: %f",3.14*Radius*Radius);
    getch();
}
```

Output :

A screenshot of a terminal window with a black background and white text. The text shows the program's execution: it prompts for the radius, receives the input '3', and then displays the calculated area of the circle as 28.260000 followed by a cursor underscore.

```
Enter Radius Of Circle:
3
The Area Of Circle is: 28.260000_
```

(5) Write a program that input two values from user and find out Area of Rectangle.

Algorithm :

Step 1 : START

Step 2 : DECLARE variable l, b, area

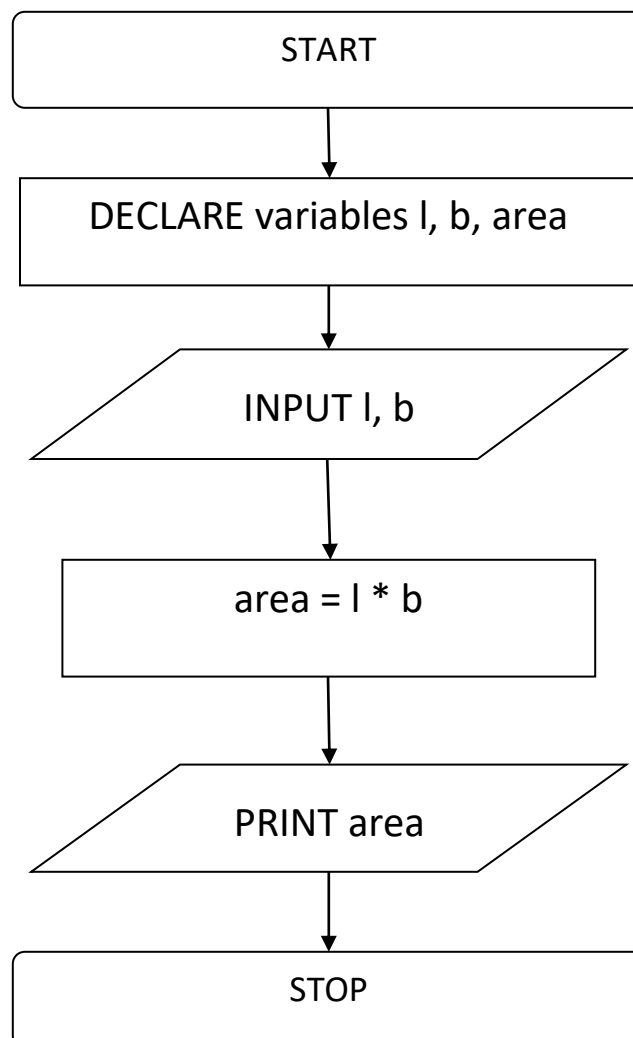
Step 3 : INPUT l, b

Step 4 : $\text{area} = l * b$

Step 5 : PRINT area

Step 6 : STOP

Flowchart :

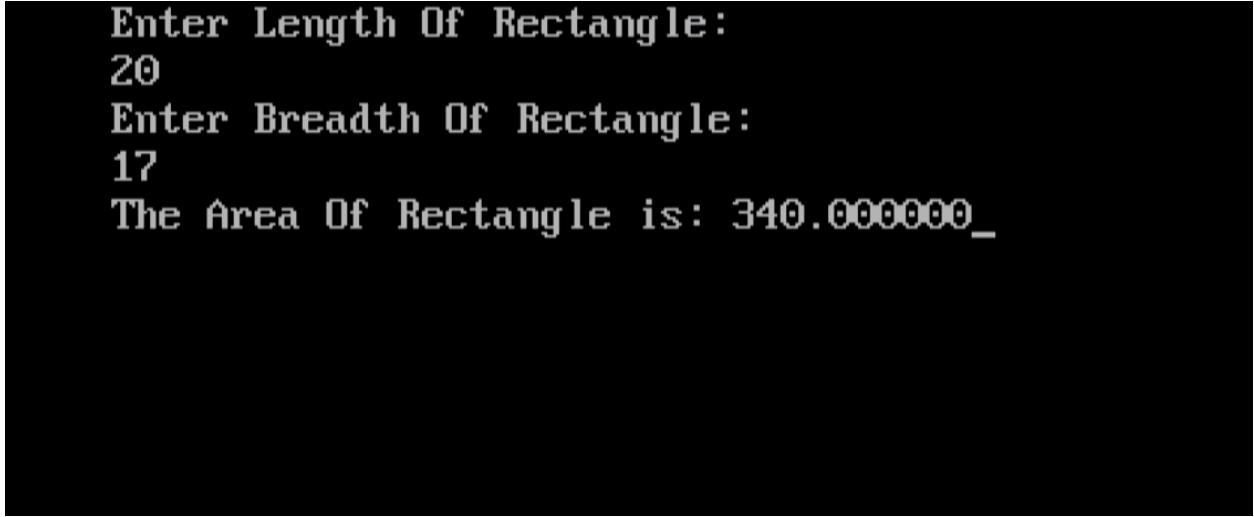


Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    float Length,Breadth;
    clrscr();
    printf("Enter Length Of Rectangle: \n");
    scanf("%f",&Length);
    printf("Enter Breadth Of Rectangle: \n");
    scanf("%f",&Breadth);
    printf("The Area Of Rectangle is: %f",Length*Breadth);
    getch();
}
```

Output :



```
Enter Length Of Rectangle:
20
Enter Breadth Of Rectangle:
17
The Area Of Rectangle is: 340.000000_
```

(6) Write a program that input 3 values from user and find out its Average.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables x, y, z, avg

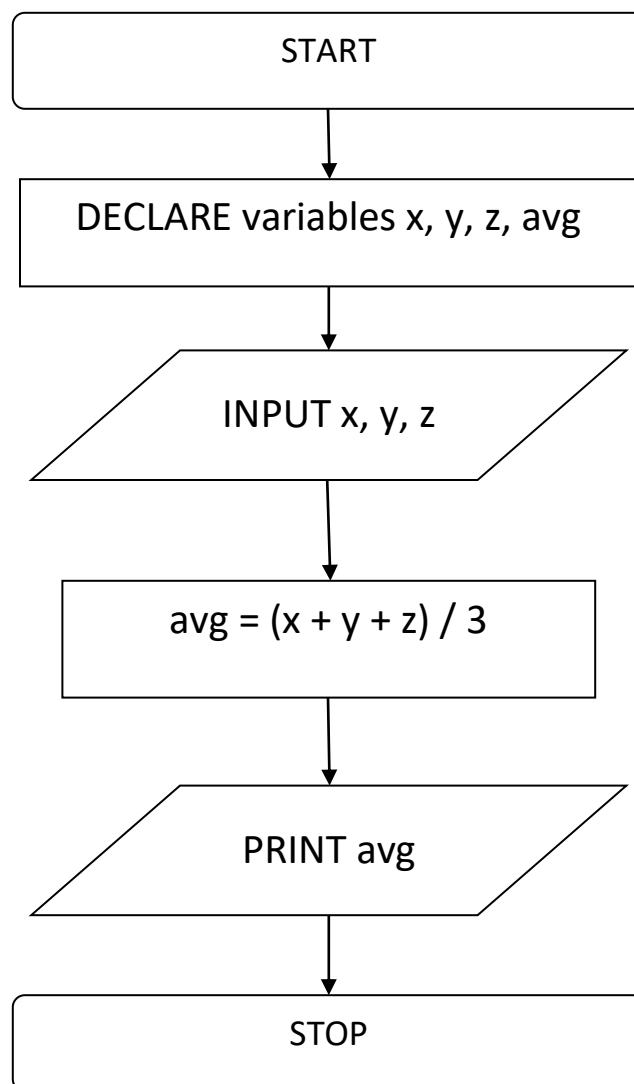
Step 3 : INPUT x, y, z

Step 4 : $\text{avg} = (x + y + z) / 3$

Step 5 : PRINT avg

Step 6 : STOP

Flowchart :

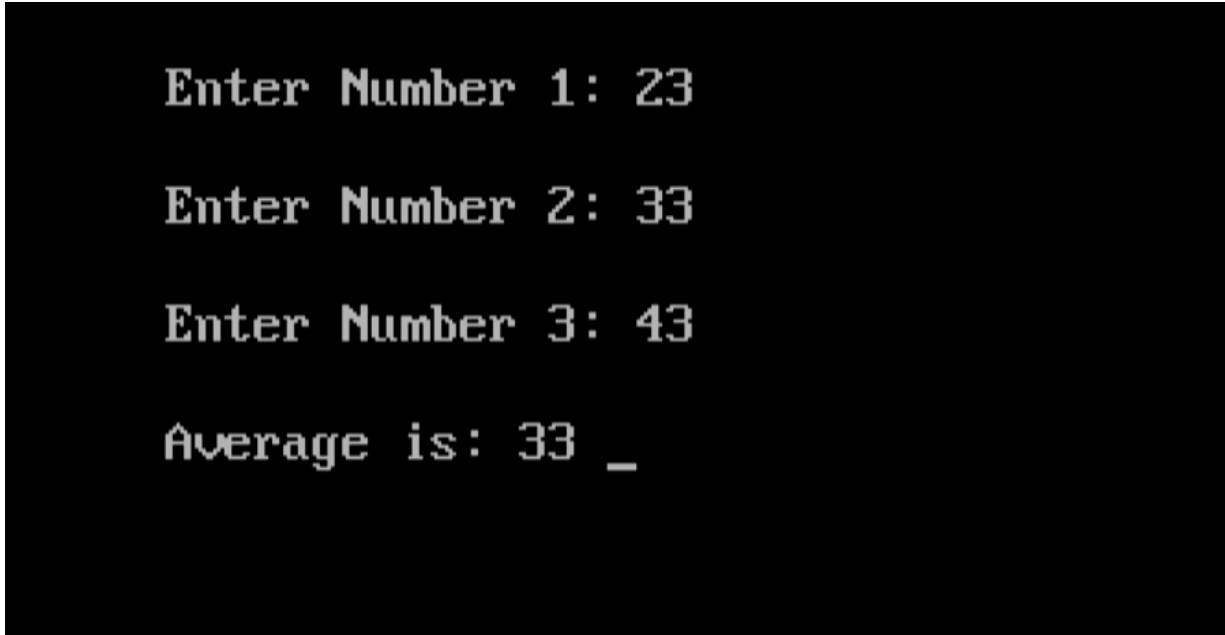


Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int x,y,z;
    clrscr();
    printf("\n Enter Number 1: ");
    scanf("%d",&x);
    printf("\n Enter Number 2: ");
    scanf("%d",&y);
    printf("\n Enter Number 3: ");
    scanf("%d",&z);
    printf("\n Average is: %d ",(x+y+z)/3);
    getch();
}
```

Output :



```
Enter Number 1: 23

Enter Number 2: 33

Enter Number 3: 43

Average is: 33 _
```

(7) Write a program to Calculate the Square and Cube of a number.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables n, sqr, cube

Step 3 : INPUT n

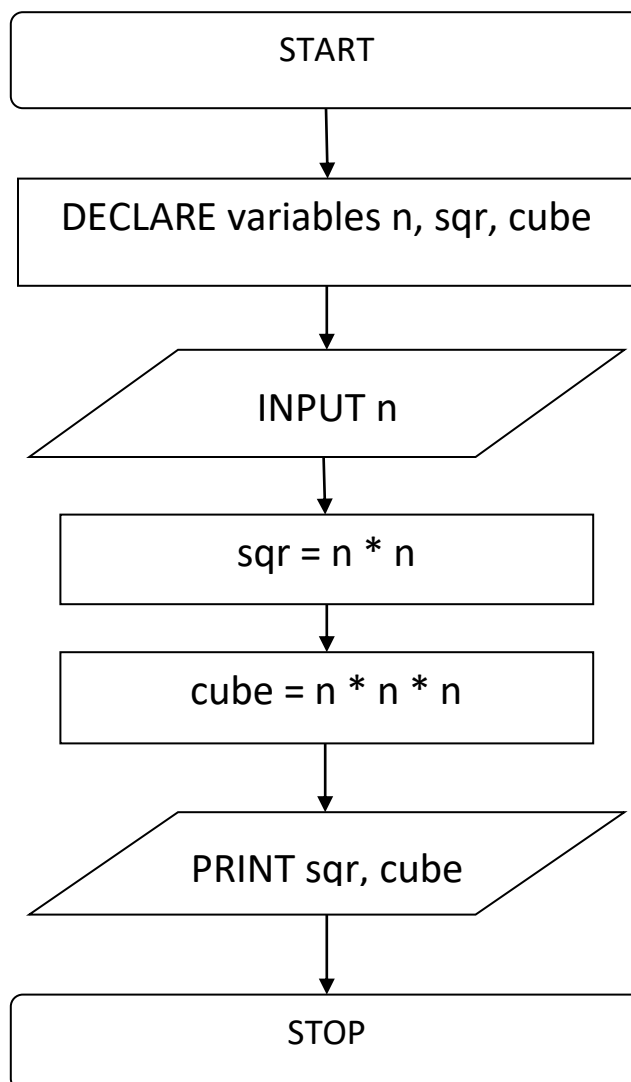
Step 4 : $\text{sqr} = n * n$

Step 5 : $\text{cube} = n * n * n$

Step 6 : OUTPUT sqr, cube

Step 7 : STOP

Flowchart :

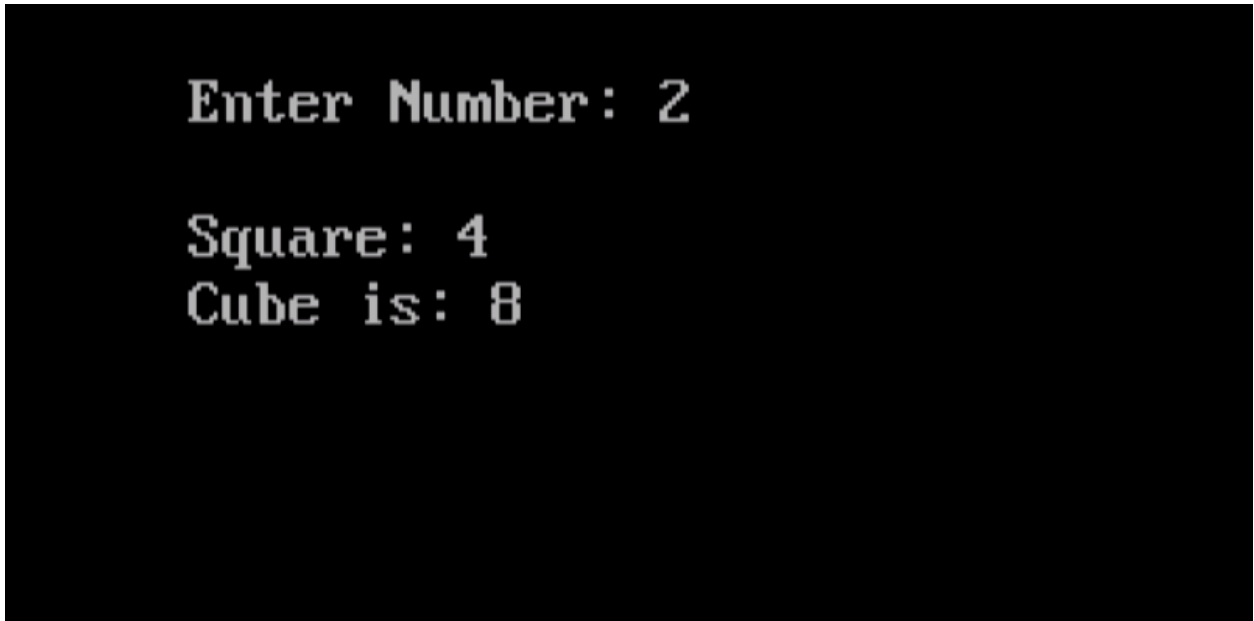


Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int n;
    clrscr();
    printf("\n Enter Number: ");
    scanf("%d",&n);
    printf("\n Square is: %d",n*n);
    printf("\n Cube is: %d",n*n*n);
    getch();
}
```

Output :



```
Enter Number: 2
```

```
Square: 4
```

```
Cube is: 8
```

(8) Write a program Interchange the value of two numbers using third variable.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables temp, x, y

Step 3 : INPUT x, y

Step 4 : temp = x

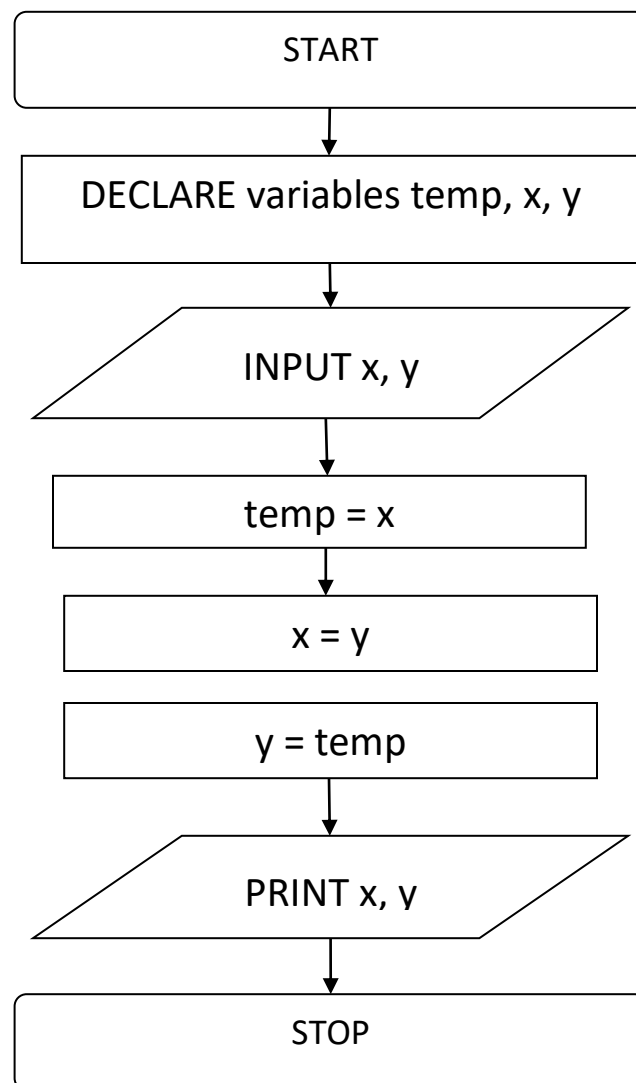
Step 5 : x = y

Step 6 : y = temp

Step 7 : PRINT x, y

Step 8 : STOP

Flowchart :

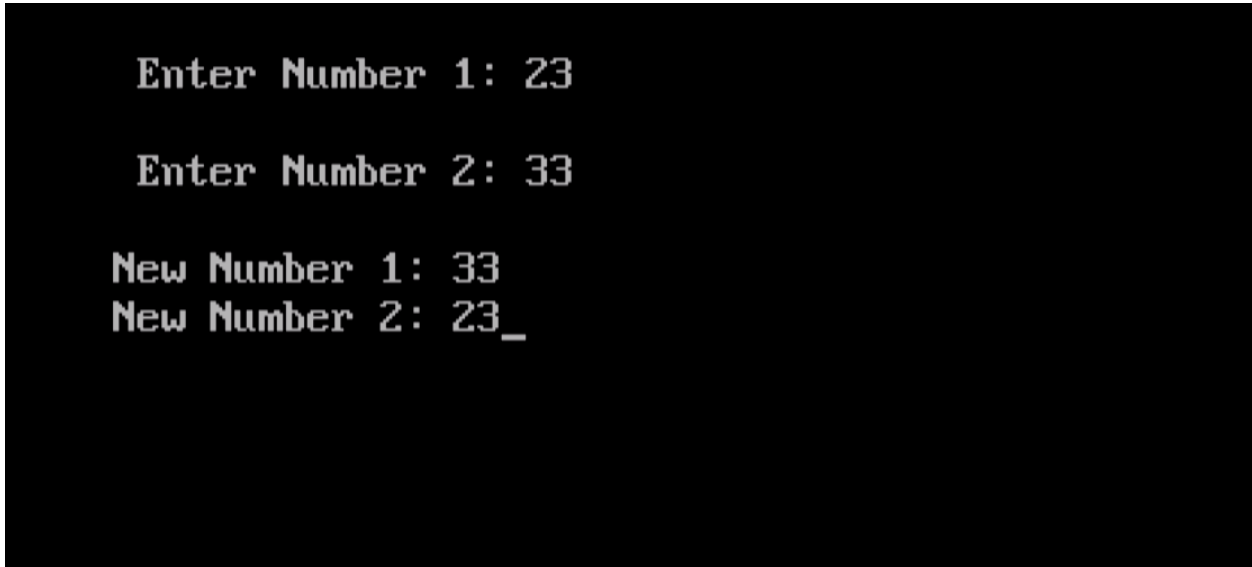


Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int temp,x,y;
    clrscr();
    printf("\n Enter Number 1: ");
    scanf("%d",&x);
    printf("\n Enter Number 2: ");
    scanf("%d",&y);
    temp=x;
    x=y;
    y=temp;
    printf("\nNew Number 1: %d",x);
    printf("\nNew Number 2: %d",y);
    getch();
}
```

Output :

A screenshot of a terminal window with a black background and white text. The output shows the program's execution: it prompts for 'Enter Number 1:' and receives '23', then prompts for 'Enter Number 2:' and receives '33'. It then displays 'New Number 1: 33' and 'New Number 2: 23_'.

```
Enter Number 1: 23

Enter Number 2: 33

New Number 1: 33
New Number 2: 23_
```

(9) Write a program Interchange the value of two numbers without using third variable.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables x, y

Step 3 : INPUT x, y

Step 4 : $x = x + y$

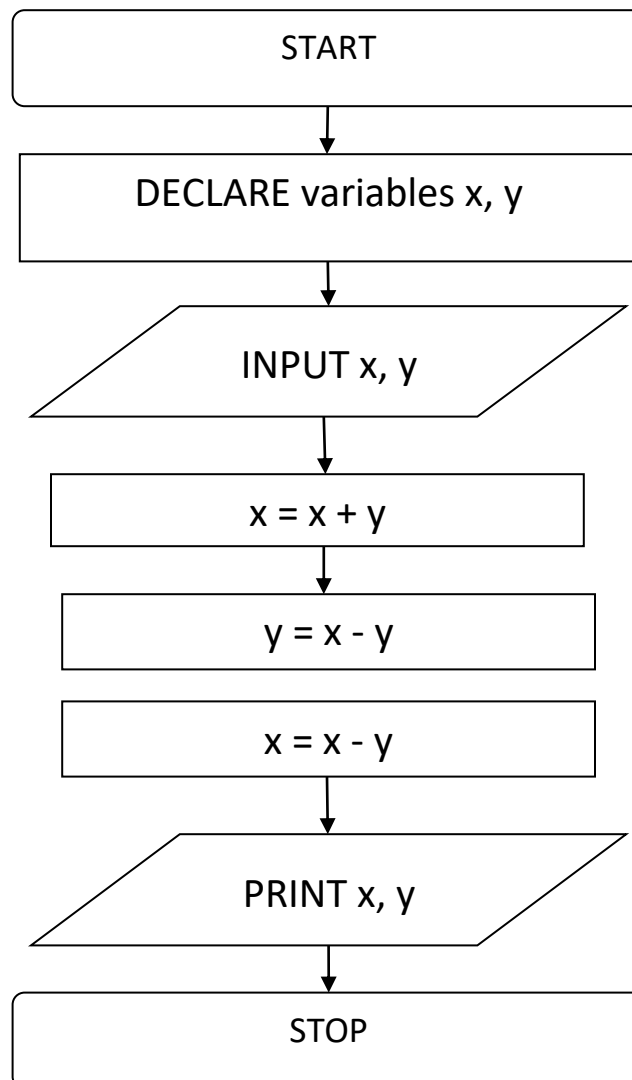
Step 5 : $y = x - y$

Step 6 : $x = x - y$

Step 7 : PRINT x, y

Step 8 : STOP

Flowchart :

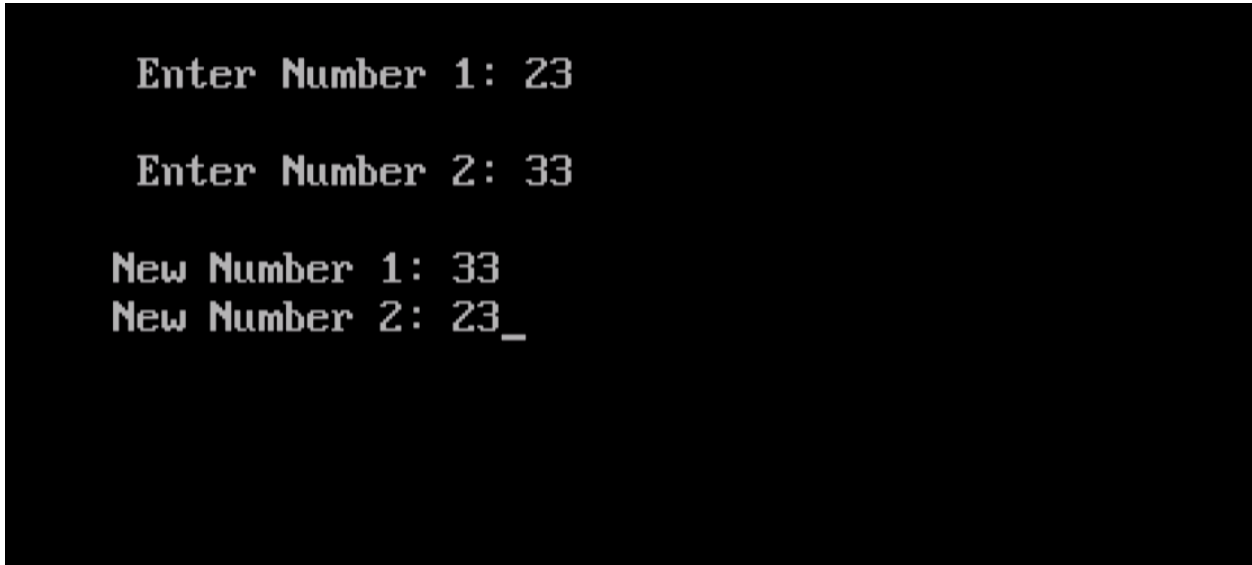


Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int x,y;
    clrscr();
    printf("\n Enter Number 1: ");
    scanf("%d",&x);
    printf("\n Enter Number 2: ");
    scanf("%d",&y);
    x=x+y;
    y=x-y;
    x=x-y;
    printf("\nNew Number 1: %d",x);
    printf("\nNew Number 2: %d",y);
    getch();
}
```

Output :

A screenshot of a terminal window with a black background and white text. The text shows the program's execution: 'Enter Number 1: 23', 'Enter Number 2: 33', 'New Number 1: 33', and 'New Number 2: 23_'.

```
Enter Number 1: 23

Enter Number 2: 33

New Number 1: 33
New Number 2: 23_
```

(10) Write a program that input quantity, price, discount and display final amount.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables q, pr, disc, f_amt

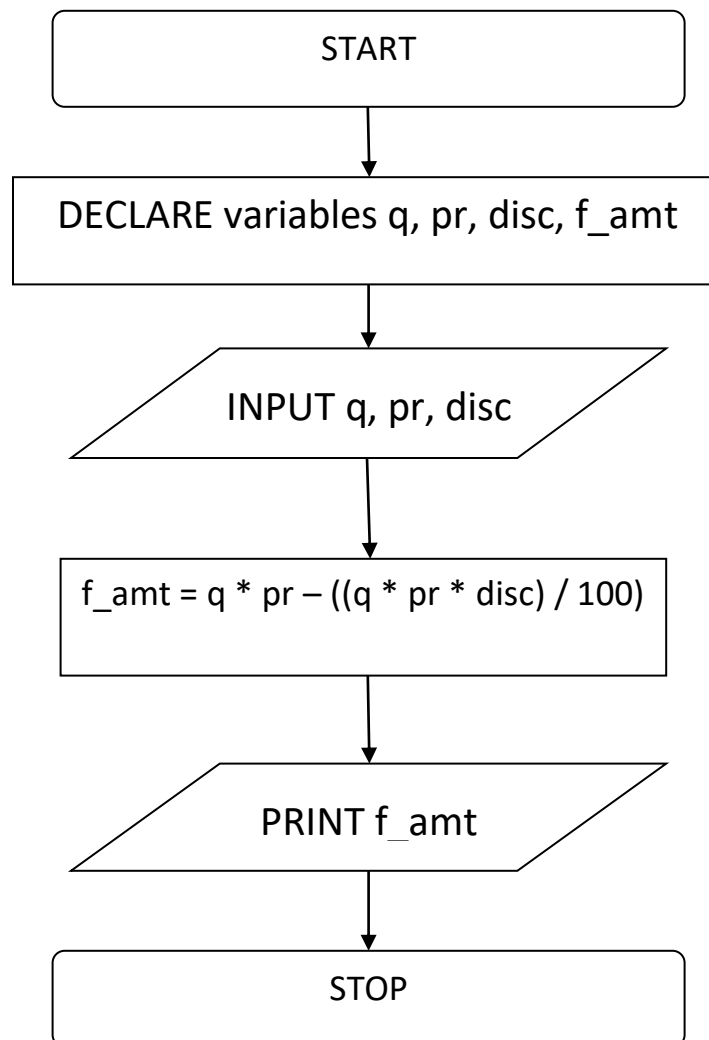
Step 3 : INPUT q, pr, disc

Step 4 : $f_amt = q * pr - ((q * pr * disc) / 100)$

Step 5 : OUTPUT f_amt

Step 6 : STOP

Flowchart :



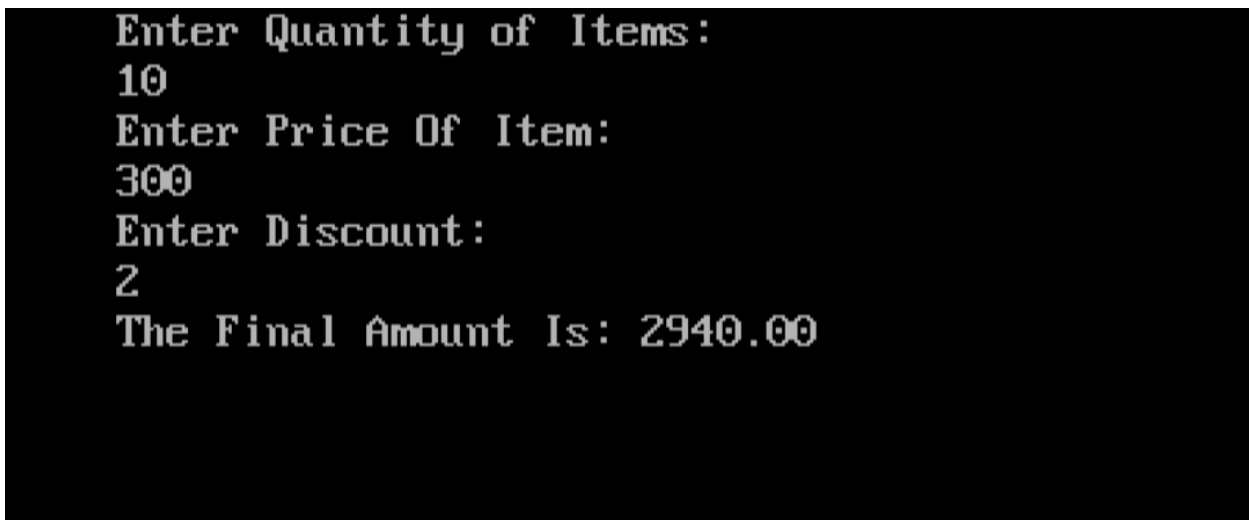
Code :

```
#include<stdio.h>
#include<conio.h>

void main(){

    float Input_quantity,Price,Discount,Final_Amount;
    clrscr();
    printf("Enter Quantity of Items: \n");
    scanf("%f",&Input_quantity);
    printf("Enter Price Of Item: \n");
    scanf("%f",&Price);
    printf("Enter Discount: \n");
    scanf("%f",&Discount);
    Final_Amount = Input_quantity*Price-((Input_quantity*Price*Discount)/100);
    printf("The Final Amount Is: %.2f",Final_Amount);
    getch();
}
```

Output :

A screenshot of a terminal window with a black background and white text. The text shows the program's execution: it prompts for 'Enter Quantity of Items:' and receives '10', then 'Enter Price Of Item:' and receives '300', then 'Enter Discount:' and receives '2'. Finally, it displays 'The Final Amount Is: 2940.00'.

```
Enter Quantity of Items:
10
Enter Price Of Item:
300
Enter Discount:
2
The Final Amount Is: 2940.00
```